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St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore–607001
B. Sc Degree Examination - APRIL 2021

APH401T – ALLIED PHYSICS

Time : 3 hrs **Max Marks :75**

SECTION – A (10X2=20)

Answer **ALL** Questions

1. What is a torsional pendulum?
2. What is transverse wave?
3. What is a capacitor?
4. What is LCR circuit?
5. Define air wedge.
6. What is optical activity?
7. What are the postulates of wave mechanics?
8. State Heisenberg's uncertainty principle.
9. What is a diode?
10. Give any two characteristics of FET.

SECTION – B (5X5=25)

Answer **ALL** Questions

11. a) State and explain the laws of transverse vibrations.

(or)

b) Explain the production and uses of ultrasonic waves.

12. a) Discuss about the loss of energy due to sharing of charges in DC circuits.

(or)

b) Explain the measurement of internal resistance of a cell using potentiometer.

13. a) Explain the test for optical flatness of glass plate

(or)

b) Explain the terms i) Specific rotatory power and ii) Polarimeter

14. a) Explain the terms i) Addition of velocities ii) Twin paradox

(or)

b) Give an account of Heisenberg's uncertainty principle.

15. a) Discuss the characteristics and parameters of FET

(or)

b) Explain the fabrication method of VLSIC.

SECTION - C (3X10=30)

Answer any **THREE** Questions

16. Explain the transverse vibrations of a stretched string. Obtain the expression for the velocity of transverse wave.

17. Derive an expression for growth and decay of circuit containing 'L'and 'R'.

18. Explain achromatic combination of two thin lenses in contact.

19. Derive one dimensional time dependent Schrödinger's equation.

20. Explain the construction and working of JK flip flop.

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